



**NATIONAL INSTITUTE OF TECHNOLOGY
PUDUCHERRY**

EE1113 – POWER ELECTRONICS LABORATORY

MINI-PROJECT

MOSFET based Timer Circuit

Semester: *IV*

Department: *Electrical and Electronics Engineering*

Degree: *B. Tech*

Academic Year: *2025-2026*

By Batch A2:

EE24B1004 - Achyut Kuumar

EE24B1005 - A. Harshitha

EE24B1006 - Anmol Sinha

Objective:

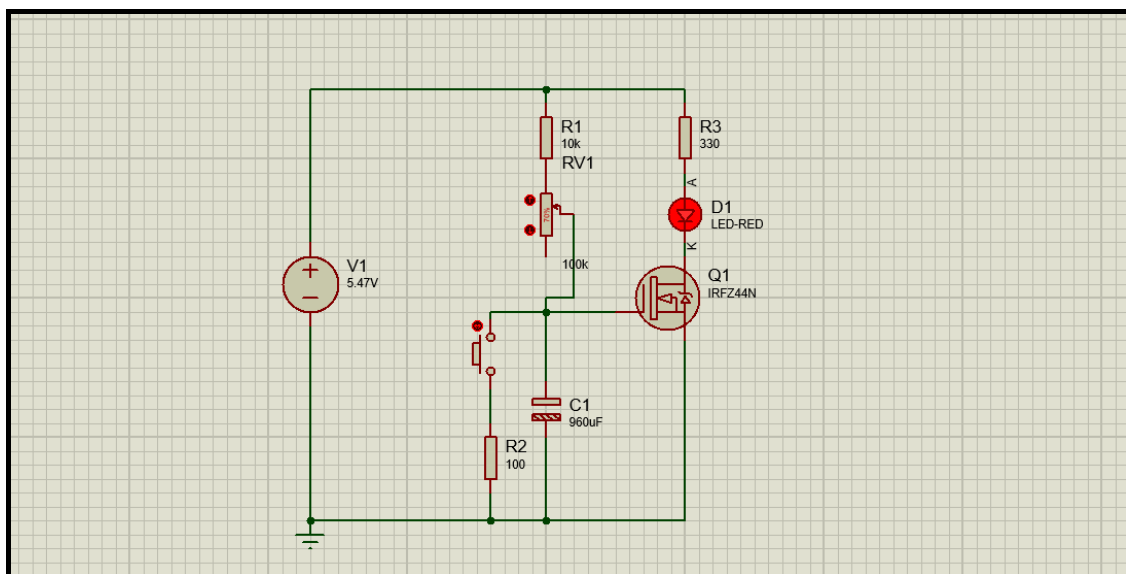
The objective of this mini project is to build an adjustable timer circuit using a resistor-capacitor timing network and an IRFZ44N MOSFET. The main goal is to control a load (an LED) so that it turns on only after a user-set time delay. An LCD is also included as a secondary feature to display this set time.

MAIN COMPONENTS:

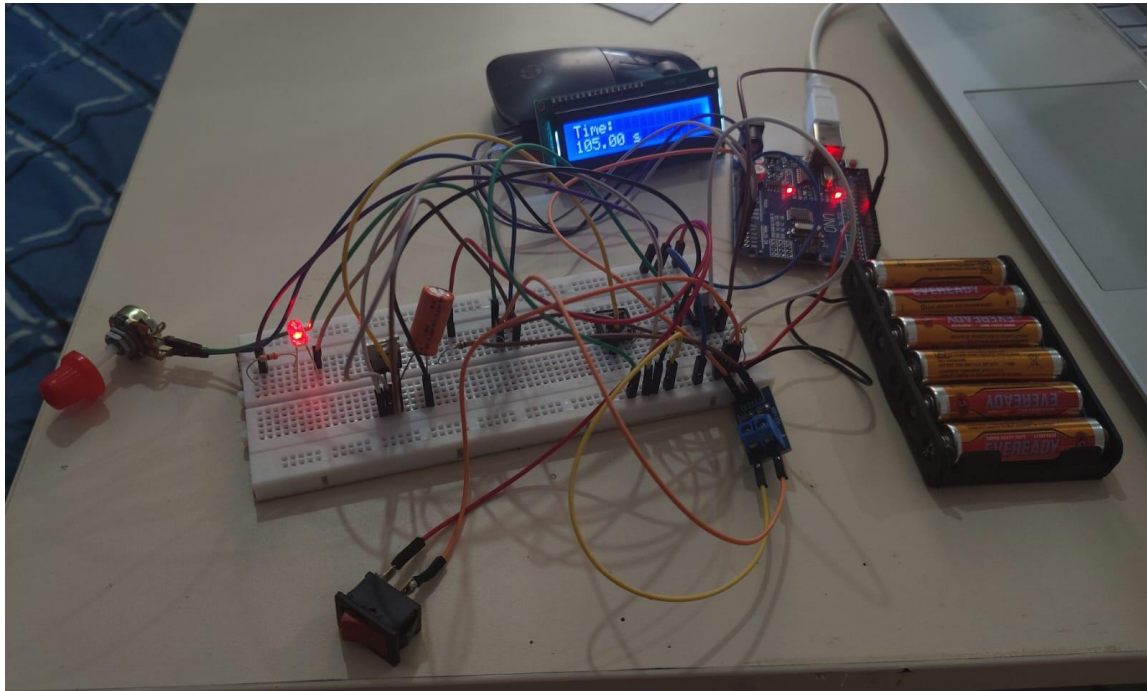
Component	Quantity
DC Supply	1
IRFZ44N MOSFET	1
LED (2V, 5V, 20mA)	1
Capacitor (960uF)	1
Breadboard	1
Voltage Sensor (0-25V)	1
Potentiometer (0-100k Ω)	5
LCD Display	1
100 Ω ,300 Ω , 10k Ω (0.25W)	1 each
Arduino Uno R4	1
Wires (Jumper and normal)	As Required

Cost: ~ 460 Rs.

CIRCUIT DIAGRAM:



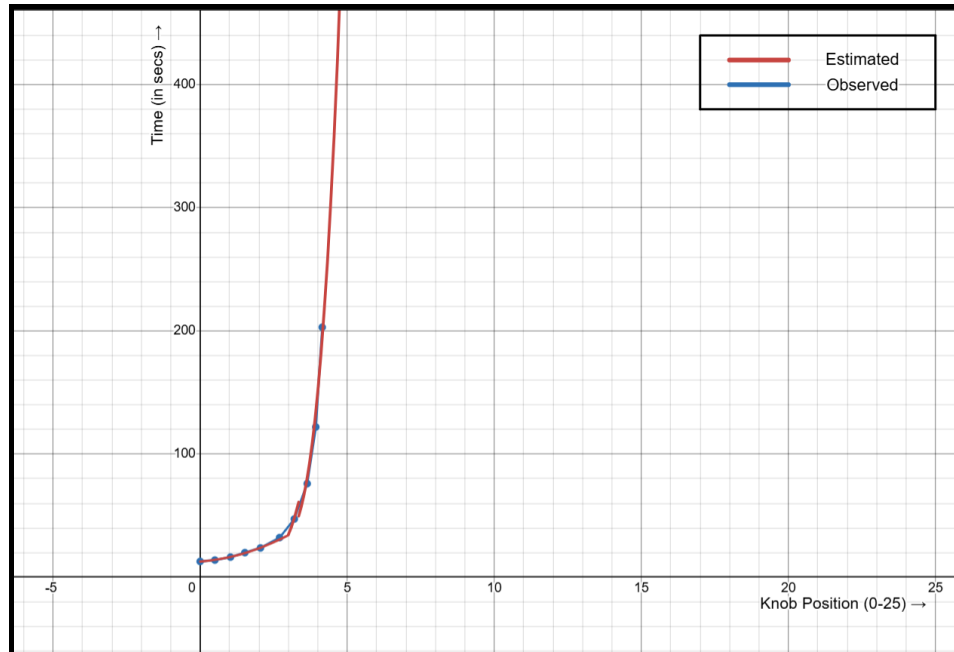
Photograph of the circuit:



Code:

```
lcdcode.ino
1  #include <LiquidCrystal_I2C.h>
2  #include <Wire.h>
3
4  float measured = 0;
5  float source = 5.47;
6  float x = 0;
7  float time = 0;
8  int button = 0;
9
10 LiquidCrystal_I2C lcd(0x27, 16, 2);
11
12 void setup() {
13   //pinMode(A0, INPUT);
14   pinMode(8, INPUT);
15   Serial.begin(9600);
16
17   lcd.init();
18   lcd.backlight();
19
20   lcd.setCursor(0, 0);      // Move cursor to character 0 on line 0
21   lcd.print("Time:");
22 }
23 void loop() {
24   measured = (analogRead(A0)/1023.0)*25;
25   x = measured;
26   if (x<3){
27     //time = 6.47646*x*x*x*x-42.30994*x*x*x+86.50814*x*x-49.85866*x+15.56221;
28     time = 1.75941*x*x+1.8666*x+12.72101;
29   }
30   else if (x<3.35) {
31     time = 10.63347*x*x*x-47.68965*x*x+57.16941*x+5.02118;
32   }
33   else{
34     time = 5.9211*x*x*x*x-38.65994*x*x*x+80.12311*x*x-46.99575*x+15.56473;
35   }
36
37   button = digitalRead(8);
38   if (button == 1){
39     lcd.setCursor(0, 1);      // Move cursor to character 0 on line 0
40     lcd.print(time);          //0 to 3.45V
41     lcd.print(" s");
42   }
43   delay(50);
44 }
45
46
```

RESULT OBTAINED:



This is the obtained plot by plotting the potentiometer knob (each position labeled from 0-25) to the time it takes from the timer to allow the flow of current. It coincides very close to the modelled timer in the code.

CONCLUSION:

The practical circuit showed that the timing graph was not completely linear. This is due to the capacitor's internal leakage current, which slightly affects its charging speed.

The LED showed a small amount of brightness just before the exact trigger time. This occurs because the MOSFET turns on gradually through a linear region, rather than acting as an instant digital switch.

Thus, it can be concluded that this timer is a reliable alternative for switching heavy loads in harsh electrical conditions, bridging the gap where digital, low-current timers are impractical.